A PROJECT TITLE

ON

<write title here and it should be minimum 8 words>

project report submitted in partial fulfilment of the requirements for the Course

DATA STRUCTURE ALGORITHM -3

BACHELOR OF TECHNOLOGY

in

A.Y.: 2026-27

by

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DEPARTMENT OF DATA STRUCTURE ALGORITHM

Name of Title: Résumé–Job Matching & Talent Marketplace Engine

1. Abstract

The project titled **"Résumé–Job Matching and Talent Marketplace Recommendation System Using Advanced Data Structures and Algorithms"** aims to develop an intelligent recruitment platform that efficiently connects job seekers with suitable employment opportunities and assists recruiters in identifying qualified candidates. In the current recruitment landscape, organizations receive a large number of applications for job openings, making the screening and selection process time-consuming and inefficient. Similarly, job seekers often face difficulties in finding opportunities that closely match their skills, qualifications, and career interests. These challenges highlight the need for an automated and efficient matching system.

The proposed system enables candidates to create profiles, upload résumés, search for available jobs, and receive personalized job recommendations. Recruiters can post job requirements and obtain ranked lists of potential candidates based on their suitability for the position. The primary objective of the system is to improve recruitment efficiency through intelligent matching and recommendation mechanisms.

The project emphasizes the application of **Data Structures and Algorithms (DSA)** rather than basic data management operations. Various DSA concepts are incorporated to enhance performance and scalability. **Hashing** is used for efficient skill indexing and retrieval; **searching algorithms** facilitate quick access to relevant jobs and candidate profiles, and **sorting techniques** help rank results according to matching scores. **Heaps and priority queues** are utilized to generate top candidate and job recommendations, while **trying** to support fast skill and job-title searches with autocomplete functionality. Additionally, **graph-based relationships** can be used to represent connections between candidates, skills, and job postings for improved recommendation accuracy.

The proposed solution demonstrates how DSA concepts can be applied to solve real-world recruitment challenges. By providing efficient searching, ranking, and recommendation capabilities, the system reduces manual effort, improves matching accuracy, and enhances the overall hiring experience for both candidates and recruiters. The project serves as a practical application of Data Structures and Algorithms in the domain of talent acquisition and online recruitment systems.

1. Introduction

1. The rapid growth of online recruitment and talent marketplace platforms has transformed the way organizations hire employees and how job seekers search for career opportunities. With thousands of job postings and candidate profiles available online, finding the right match has become increasingly challenging for both recruiters and applicants.

2. Recruiters often spend considerable time reviewing numerous applications, while candidates struggle to identify positions that best match their skills and qualifications. These challenges highlight the need for automated systems capable of providing accurate recommendations and efficient search capabilities.

3. The proposed **Résumé–Job Matching & Talent Marketplace Engine** addresses these issues by enabling candidates to upload résumés, search for jobs, and receive ranked recommendations. Recruiters can post job requirements and obtain ranked lists of suitable candidates. The system focuses on the effective use of **Data Structures and Algorithms** to improve performance and scalability. Hashing is used for efficient skill indexing; searching algorithms help retrieve relevant records quickly, sorting and ranking techniques prioritize results, heaps support top recommendations and tries to enable fast skill and job-title searches.

4. Objectives of the Proposed System

5. To develop an efficient talent marketplace platform for candidates and recruiters.
6. To automate résumé-job matching based on skills and qualifications.
7. To provide ranked job recommendations for candidates.
8. To generate ranked candidate recommendations for recruiters.
9. To apply Data Structures and Algorithms for efficient searching, matching, and ranking operations.
10. To reduce recruitment time and improve matching accuracy.

Signature of the Guide **Course Instructor**